Lecture 8: The Class NP

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Recall: Time Complexity

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Definition: Let M be a TM that halts on all inputs (decider). The running time or time complexity of M is the function $t\colon \mathbb{N} \to \mathbb{N}$ where

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Definition: Time complexity class

$$TIME(t(n)) = \{L \subseteq \Sigma^* \mid L \text{ is decided by a TM with running time } O(t(n))\}$$

The class P

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Some languages in P:

- $\{\langle A \rangle \ : \ A$ is a sorted array of integers$\}$
- $\{\langle G, s, t \rangle \ : \ s$ and t are vertices connected in graph $G\}$ (Breadth-First Search)
- $\{\langle G \rangle \ : \ G$ is a connected graph$\}$

NP: Easy-to-verify problems

The class **NP**

Definition: A verifier for a language A is a TM V, where

$$A = \{w \mid \exists C \text{ s.t. } V \text{ accepts } \langle w, C \rangle\}.$$

(Here C is called a *certificate* or *witness*)
A polynomial time verifier runs in polynomial time in $|w|$.

Definition: **NP** is the class of languages with polynomial time verifiers

Why is it called **NP**?

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An NTM is a **polynomial time** NTM if the running time its longest computation on x is **polynomial in $|\mathbf{x}|$**.

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Now $x \in L$ iff there exists C such that V accepts $\langle x, C \rangle$.

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Then

$$\mathrm{NP} = \bigcup_{k=1}^{\infty} \mathrm{NTIME}(n^k)\,.$$

Example problems in **NP**

Satisfiability problem, SAT

Conjunctive Normal Form (CNF) Formula:

$$\varphi_1 = (\overline{x} \vee \overline{y} \vee z_0) \wedge (x \vee \overline{y} \vee z_1) \wedge (\overline{x} \vee y \vee z_2) \wedge (x \vee y \vee z_3)$$

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Check: φ_1 has 32 satisfying assignments, φ_2 as only one, φ_3 has zero.

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A formula is **satisfiable** if it has at least one satisfying assignment.

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$$\begin{aligned} \text{SAT} &= \{\langle\varphi\rangle \ : \ \varphi \text{ is satisfiable}\} \\ &= \{\langle\varphi\rangle \ : \ \exists C \text{ such that } C \text{ evaluates } \varphi \text{ to TRUE}\} \end{aligned}$$

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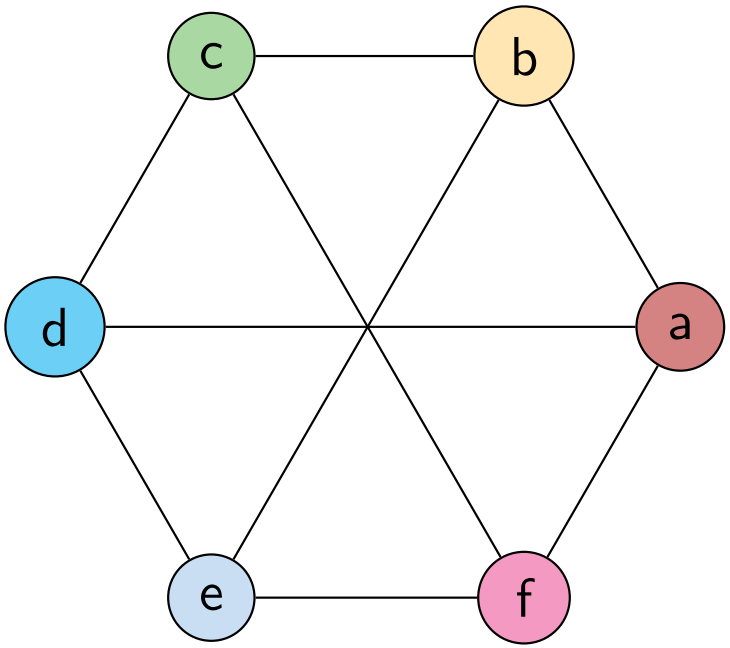
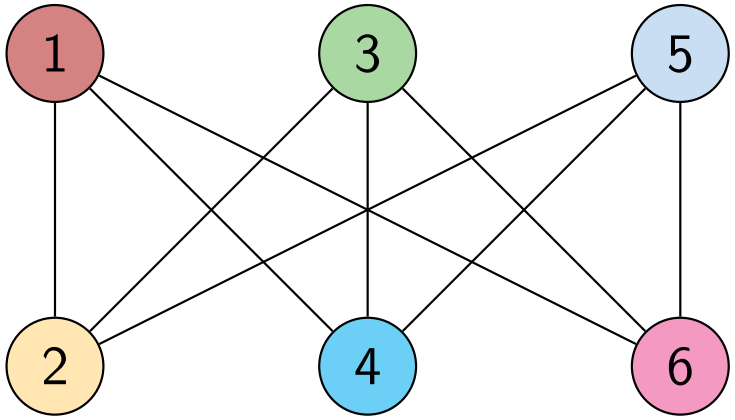
Graph isomorphism, GI





Graph Isomorphism: Bijection $f : V(G_1) \longrightarrow V(G_2)$ which preserves adjacency: $\{u, v\} \in E(G_1) \Leftrightarrow \{f(u), f(v)\} \in E(G_2)$

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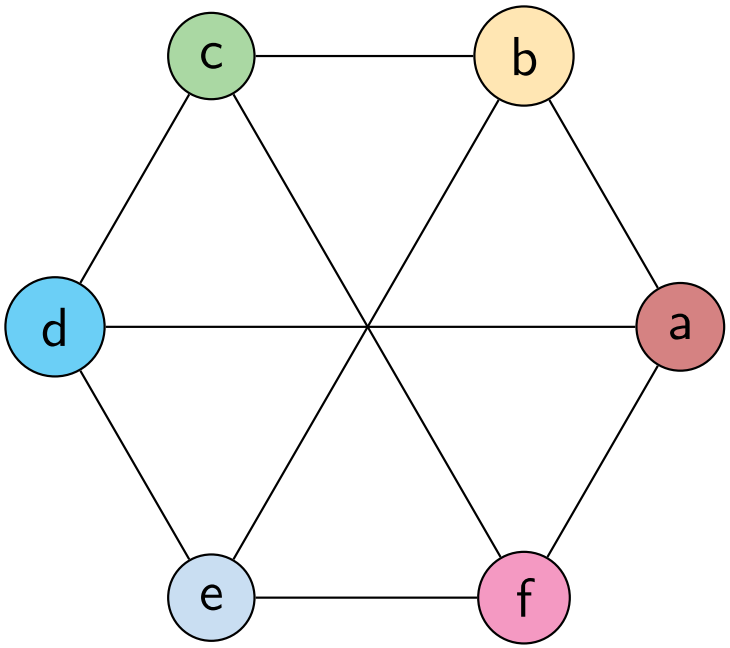

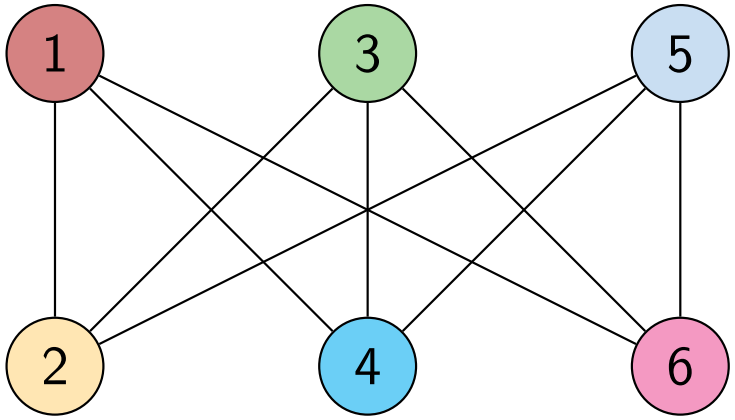

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Eg. $a \to 1$ $b \to 2$ $c \to 3$ $d \to 4$ $e \to 5$ $f \to 6$ in the graphs above.

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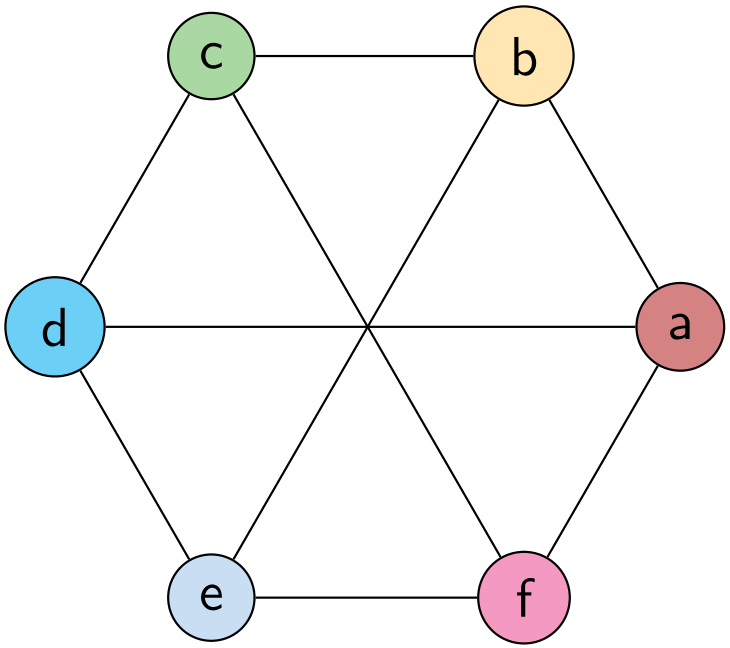

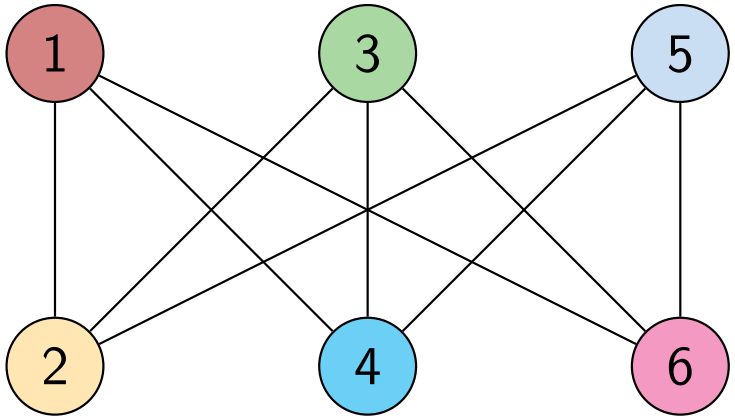

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Two graphs are **isomorphic** if they have at least one graph isomorphism.

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4. Check that C preserves adjacency, that is, for each $u, v \in V(G_1)$:
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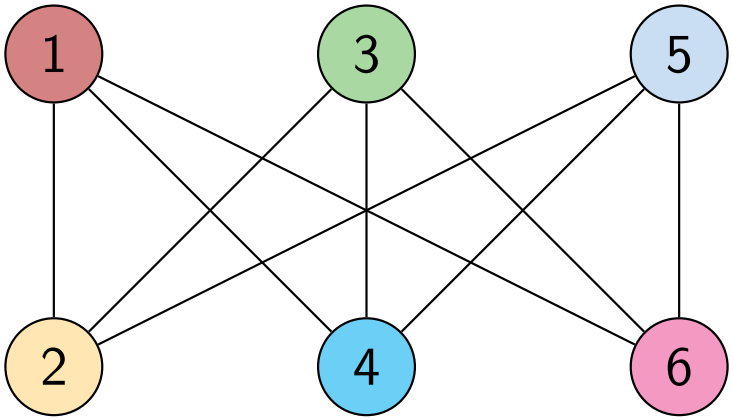
Independent set, INDSET



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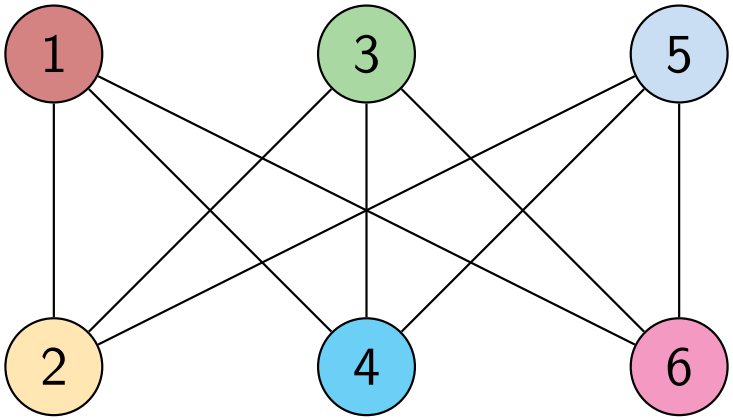

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Eg. $\{1, 3, 5\}$, $\{2, 4\}$, $\{6\}$, $\emptyset$, etc. in the graph above.

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P and NP

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Is $\mathsf{P} = \mathsf{NP}$? Nobody knows ...

Find the answer and win USD 1,000,000!

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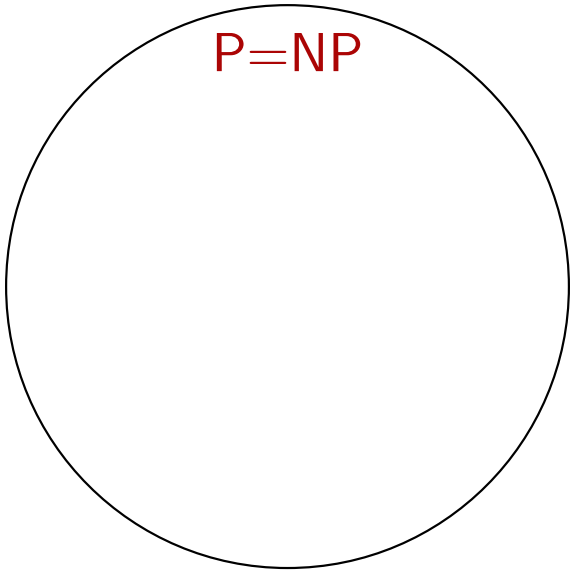

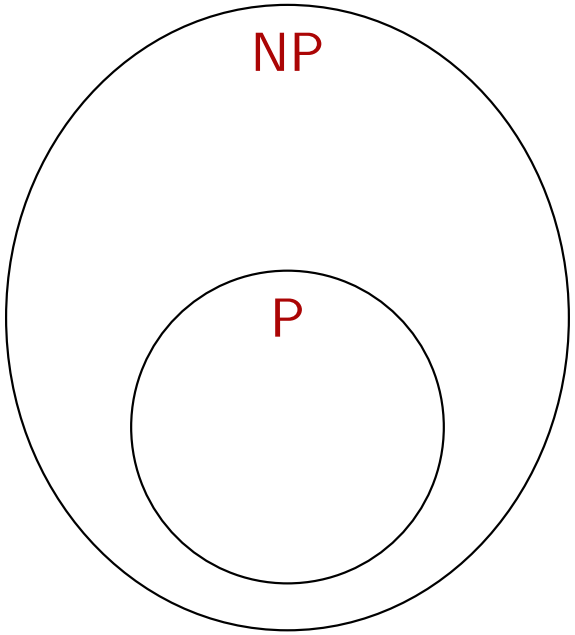

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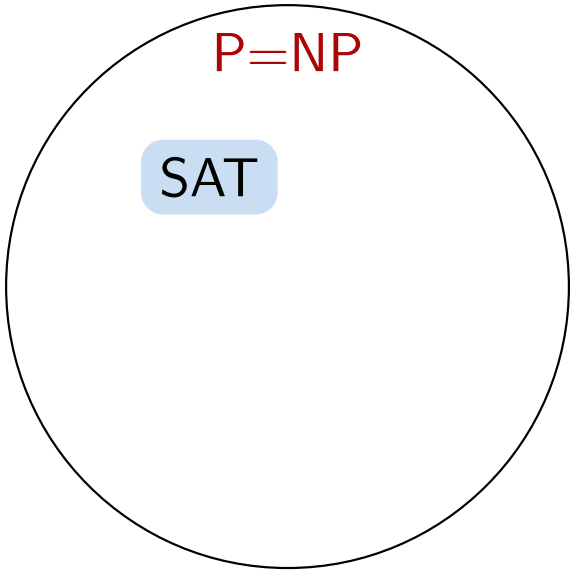

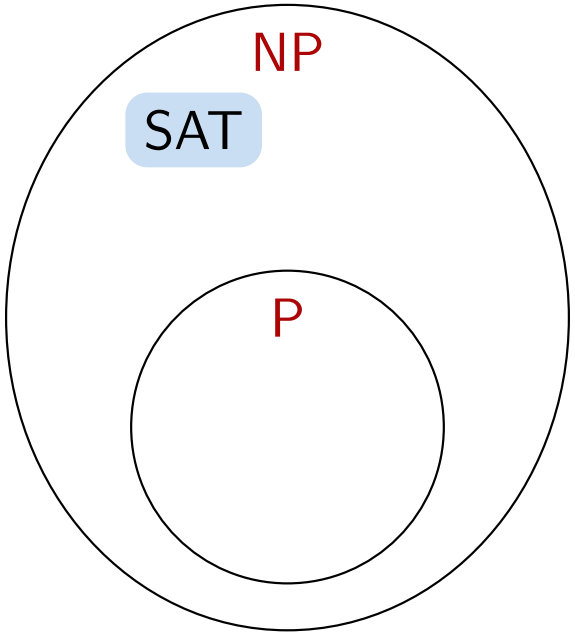

Cook–Levin Theorem (informal): $SAT \in P$ iff $P = NP$.

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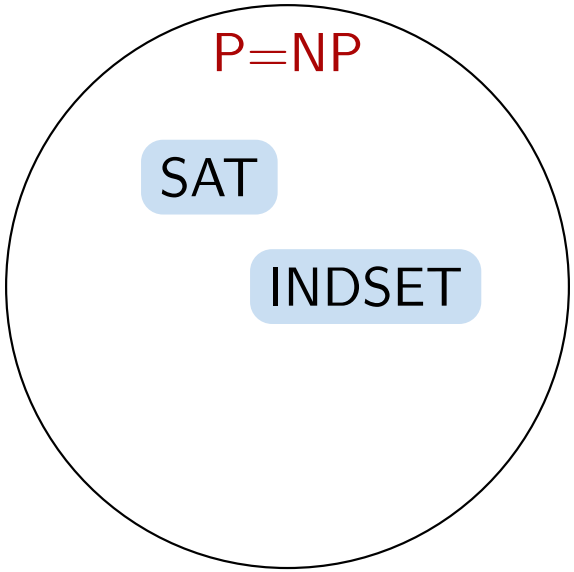

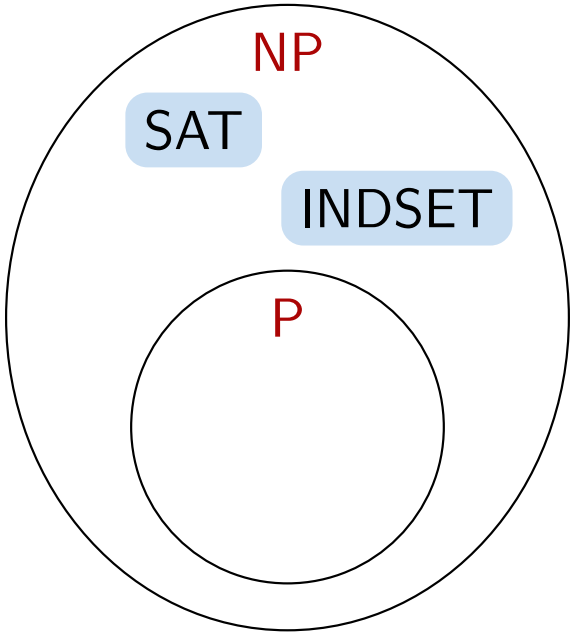

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(Also INDSET $\in$ P iff $P = NP$.)

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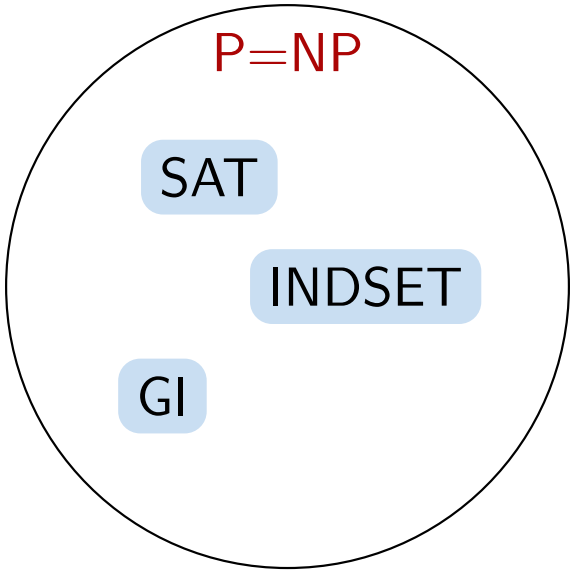

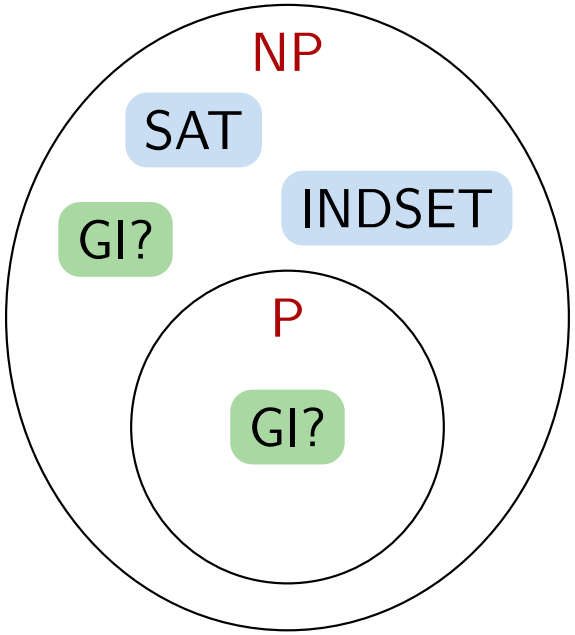

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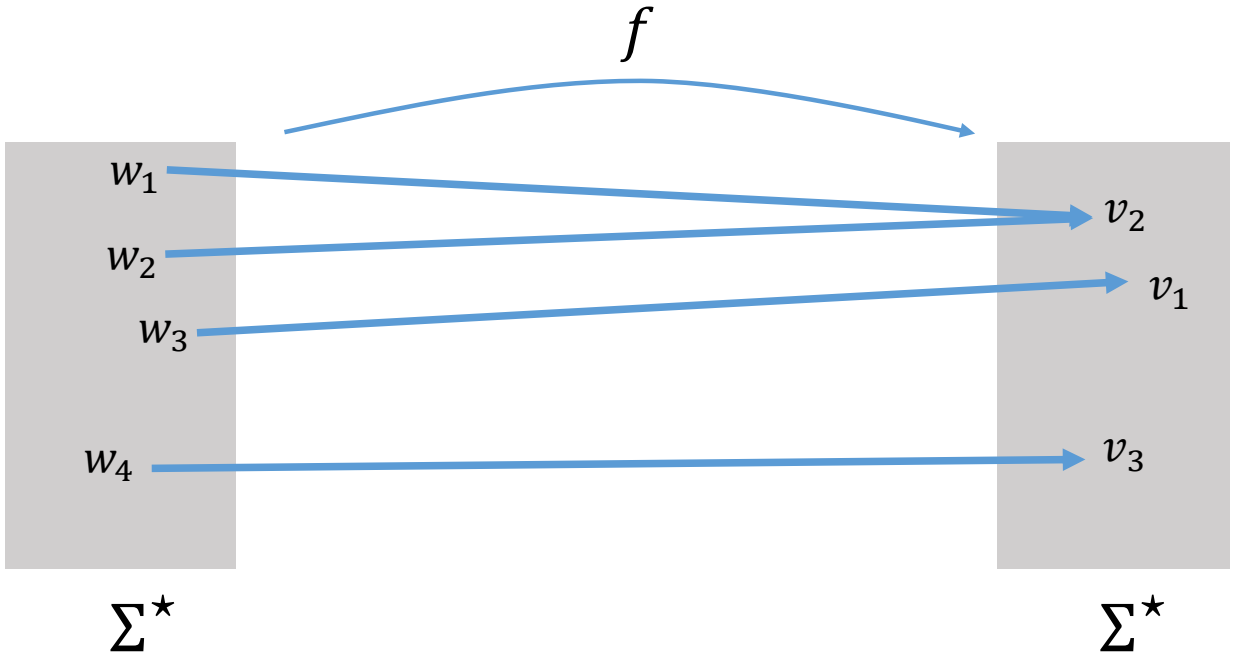
Polynomial-Time Reductions

Reductions Part 1: Poly-time Computability



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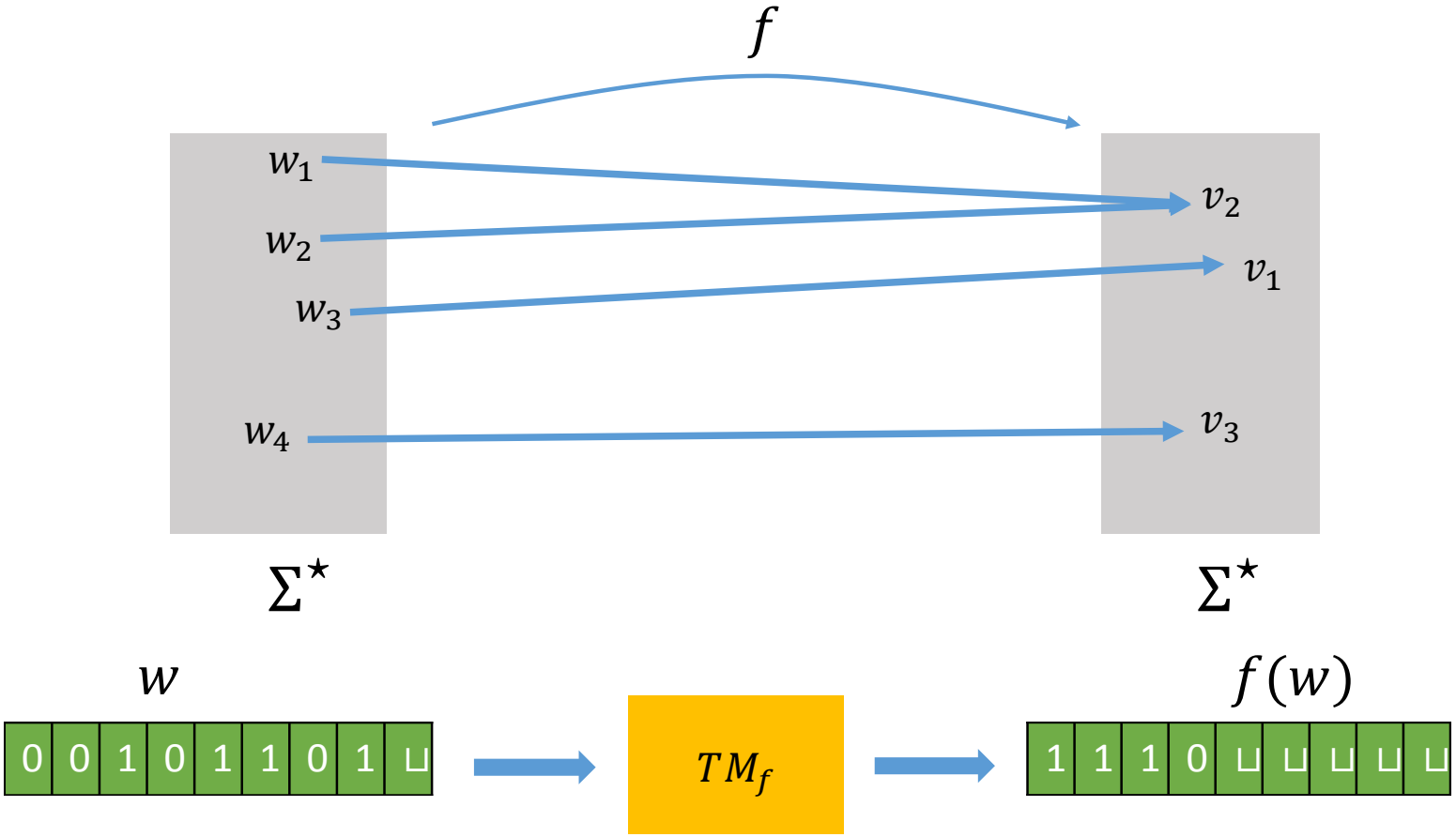
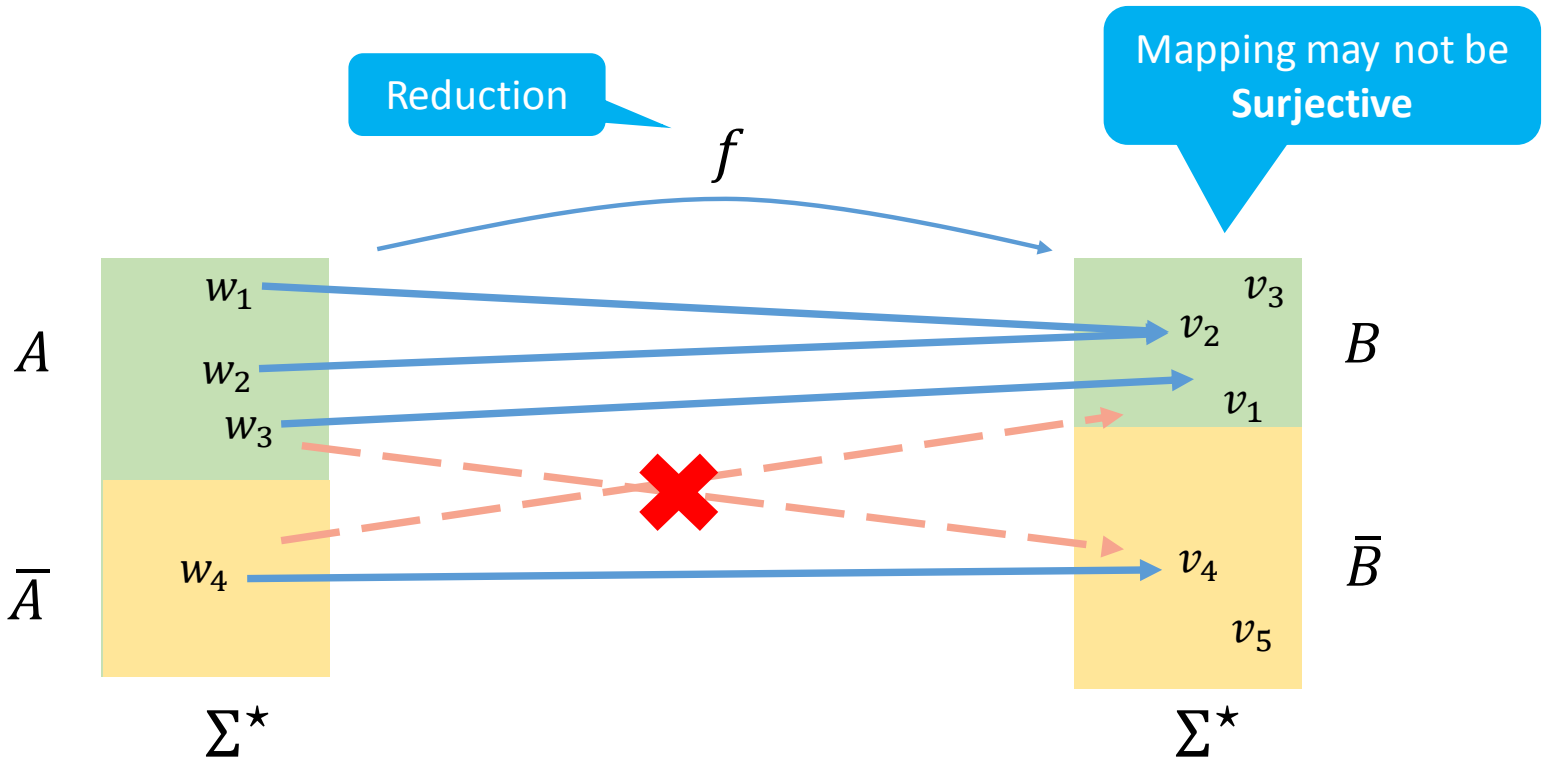


Definition: A function $f : \Sigma^* \to \Sigma^*$ is a *poly-time computable function* if some poly-time TM M, on every input w halts with just $f(w)$ on its tape.

Reductions Part 2: Correctness



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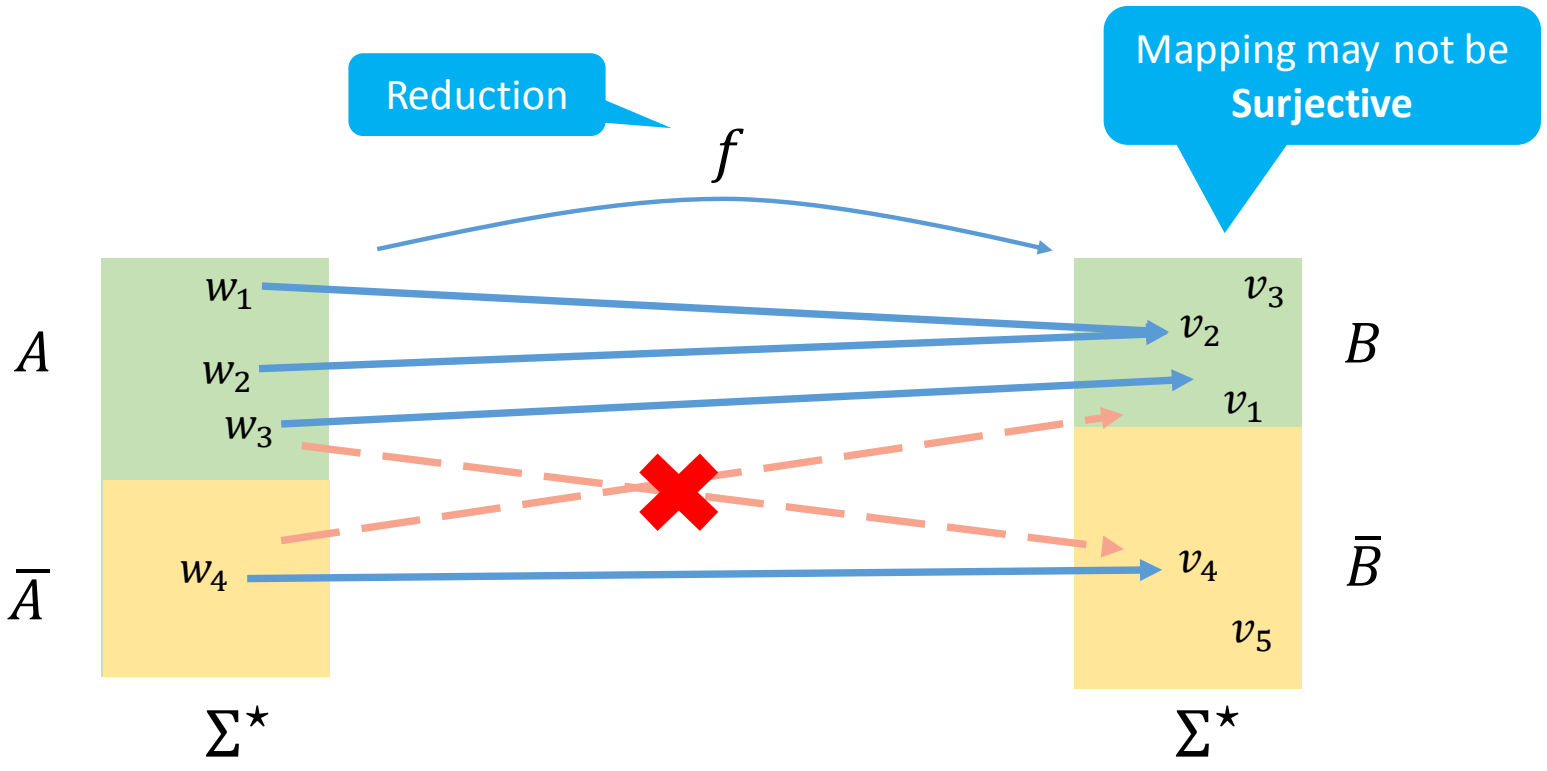

Definition: Language A is **poly-time mapping reducible** to language B, written $A \leq_P B$, if there is a poly-time computable function $f : \Sigma^* \to \Sigma^*$, such that for every $w \in \Sigma^*$:

$$w \in A \Leftrightarrow f(w) \in B$$

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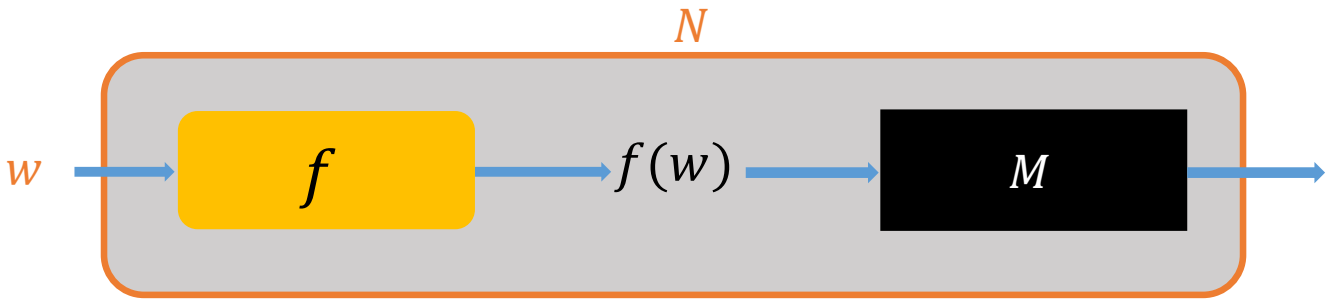

- $N =$ "On input w :
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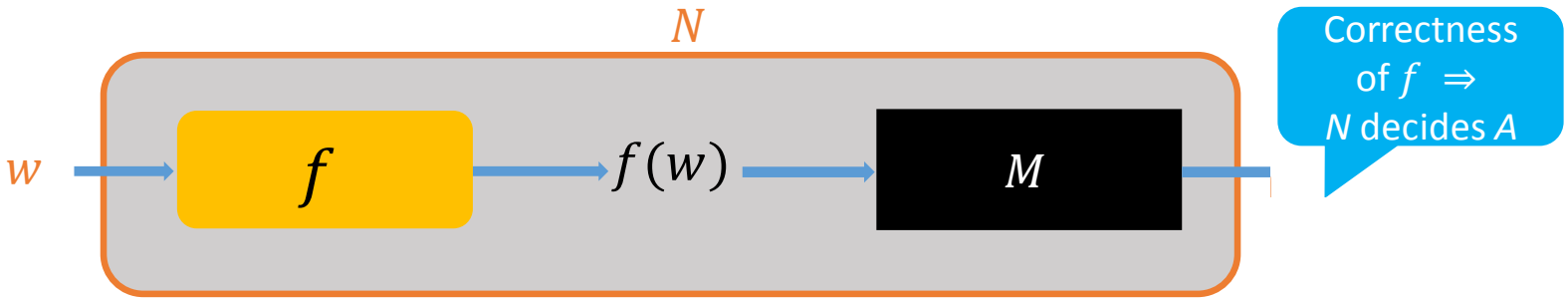

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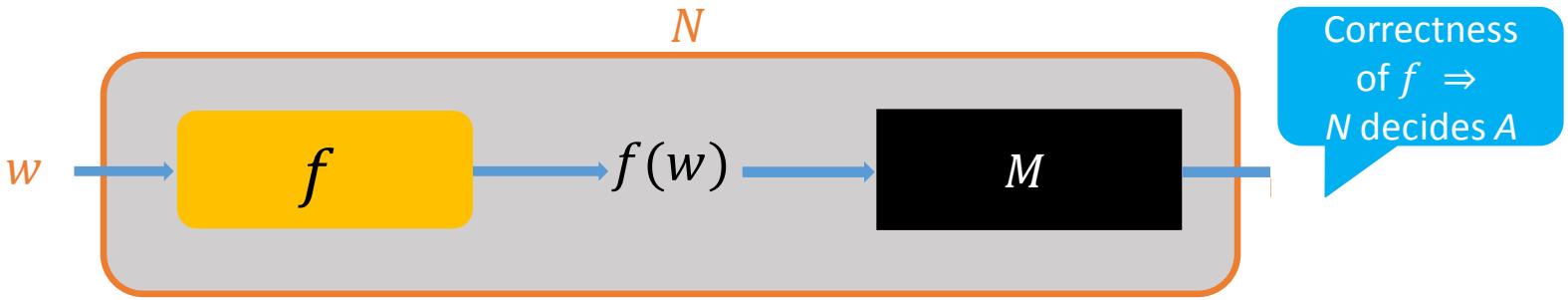

- $N =$ "On input w :
  1. Compute $f(w)$ $(O(|w|^q)$ time; $|f(w)| = O(|w|^q))$
  2. Run M on input $f(w)$ and output whatever M outputs" $(O(|w|^{pq})$ time)

Corollary: If $A \leq_P B$ and A is not in P, then B is not in P.

Theorem: If $A \leq_P B$ and $B \leq_P C$, then $A \leq_P C$. (i.e. $\leq_P$ is transitive.)

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Proof:

- Assume f_{AB} is an $O(n^p)$-time reduction from A to B
  and f_{BC} is an $O(n^q)$-time reduction from B to C.
- $N =$ "On input w :
  1. Compute $f_{AB}(w)$
  2. Compute $f_{BC}(f_{AB}(w))$"

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  2. Compute $f_{BC}(f_{AB}(w))$" $(O(|w|^{pq})$ time)
- N computes a poly-time reduction $A \leq_P C$.

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To show L is NP-complete:

- **[NP membership]** Give a poly-time verifier for L.
- **[NP hardness]** Show that SAT $\leq_P L$
  (or take any L^* already proven to be NP-complete,
  and show $L^* \leq_P L$).

Examples of NP-completeness proofs

3SAT is NP-complete

$$k\text{SAT} = \{\varphi \ : \ \varphi \text{ is satisfiable and each clause of } \varphi \text{ contains } \leq k \text{ literals}\}$$

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  Replace K with the following two clauses

$$K_1 = (\ell_1 \vee \ell_2 \vee z)$$
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Does SAT $\leq_P$ 2SAT analogously?

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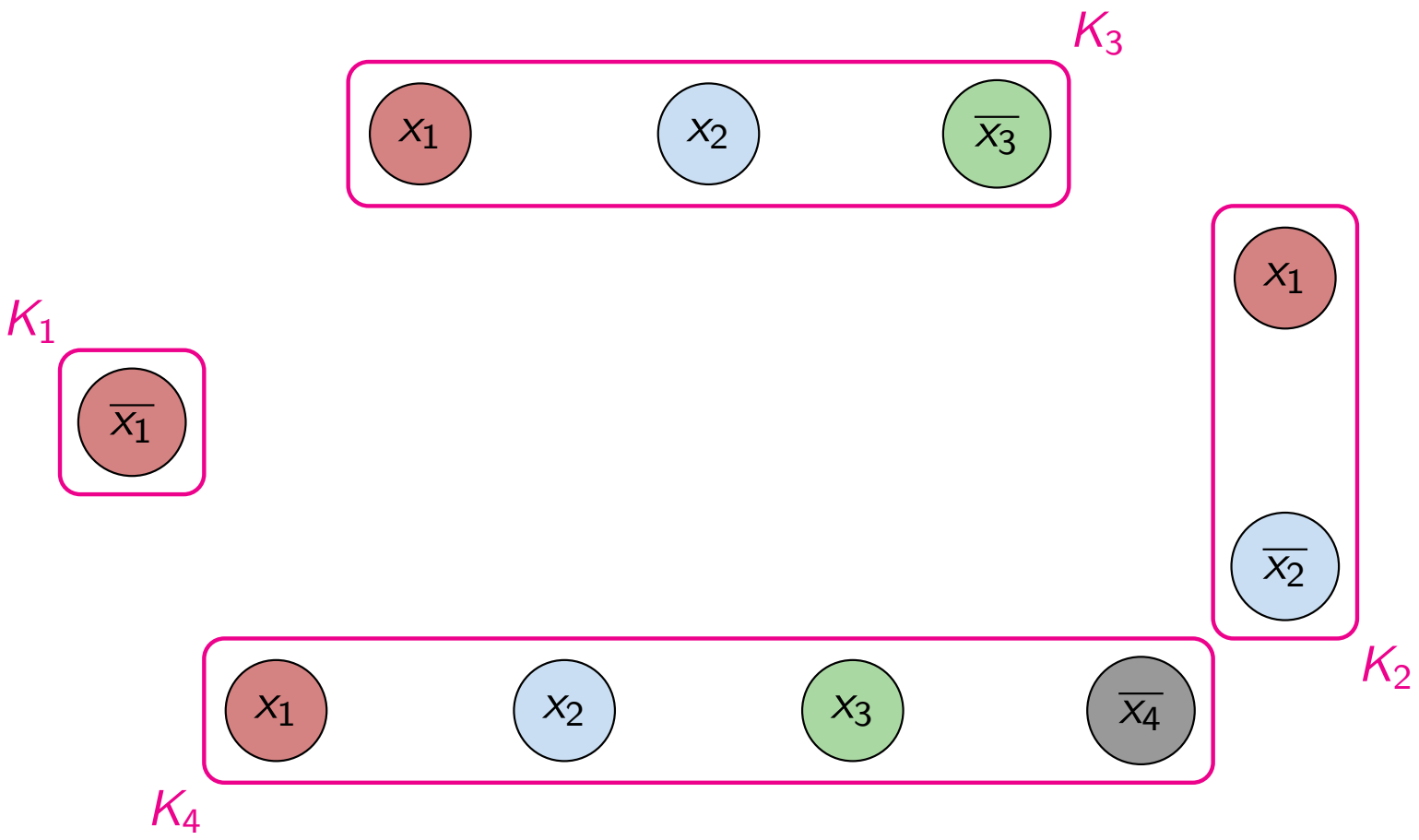

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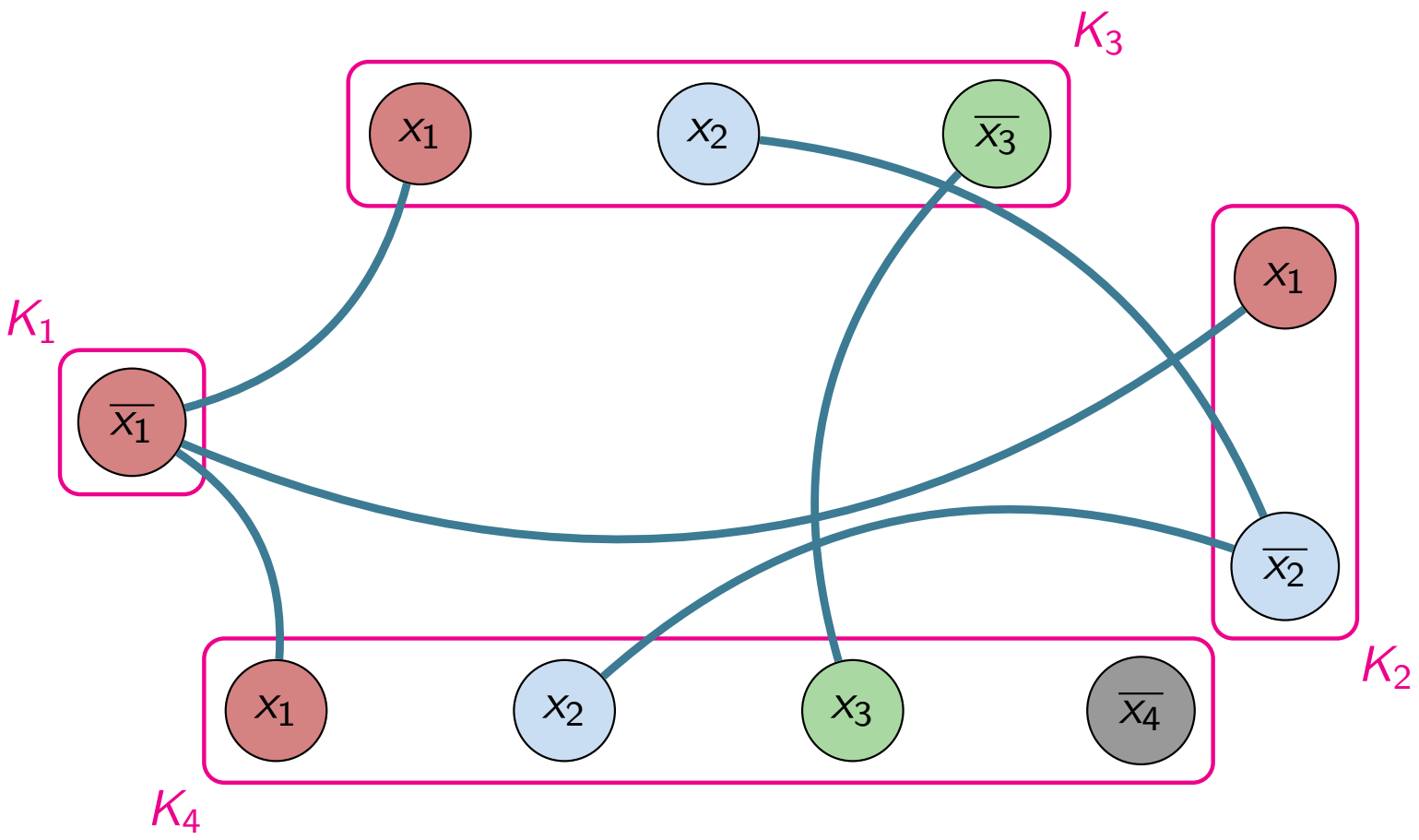

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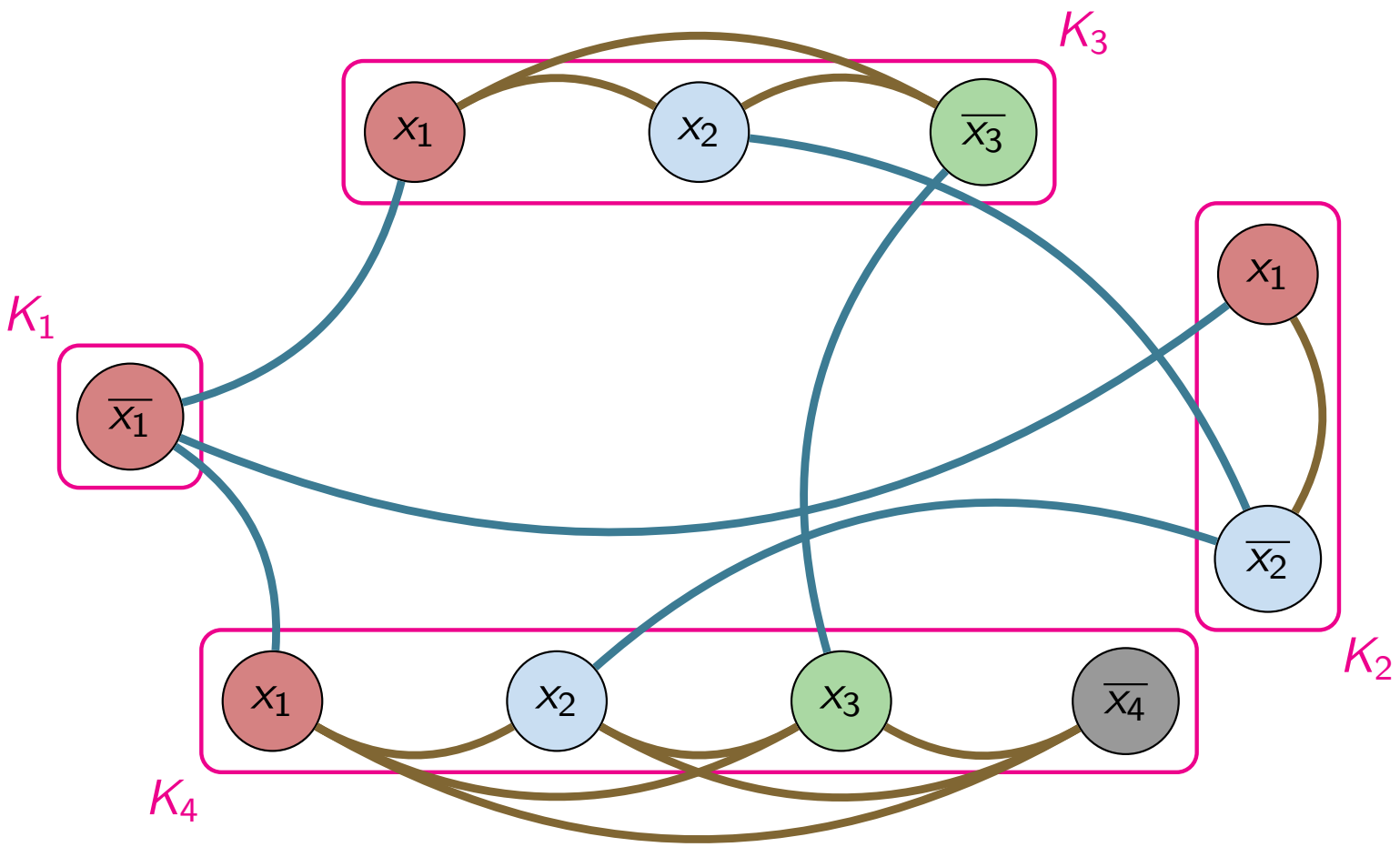

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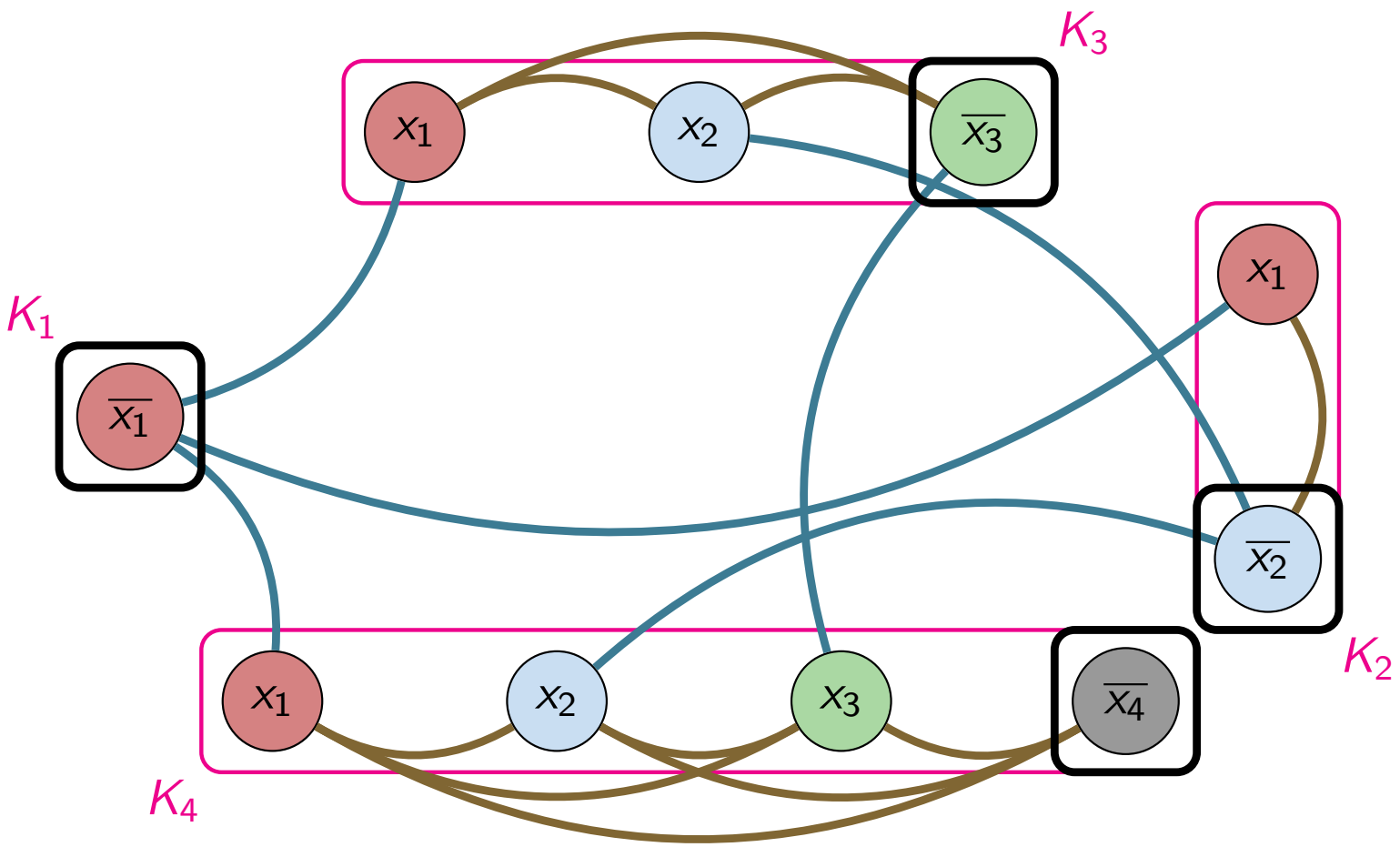

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3. Output (G, m).

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Claim: $f(\varphi) \in \text{INDSET} \implies \varphi \in \text{SAT}$

Proof: C: independent set in G, $|C| = m$. C contains one vertex from each group. Set the corresponding literals to `true` to get a satisfying assignment.

Next class

Will continue NP-completeness.